Elastoplasticity

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0.1 The Elastoplasticity Problem

0.1.1 Description and Connection with Cold Spray

Plasticity is the study of materials that undergo irreversible, permanent deformations under large loads.

Cold spray deposition results from micron sized metal particles impacting a target substrate at supersonic speeds. Plastic and thermomechanical effects are the dominant physics in cold spray deposition. Under the viscoplastic deformation and thermomechanical behaviors, the metal particles undergo shear thinnign then metallurgically binds the target substrate. Understanding the plastic processes is necessary for modeling cold spray deposition.

0.1.2 The Mathematical Problem

Plasticity is inherently an experimental science. The mathematical theories of plasticity are formulated to represent experimental findings and physical models, and such is more phenomenological. The physical models of plasticity is concerned with how plasticity occurs.

The physical material of interest will completely dictate inelastic processes. The materials of interest can vary greatly from fairly regular metals, to soil types, and even rocks and concrete. Respectively, some modes of fault: slip bands, compactification of soil or voids and shearing, and opening and closing of cracks. [4]

To describe the onset and accumulations of permanent deformation, known as yielding, we need a criterion for the transition from elastic to plastic deformation. From physical experiments, the transition from elastic to plastic may not be obvious, if such a point even exists in reality. There are then 2 main types of constitive models for plastic deformation. One assuming the existence of a yield point, and one that does not. Here we present the former. [3]

Suppose there exists a function of all internal state variables called the yield function – $F(q)$. We denote the collection of all state variables with q. We assume the following.

$$\text{if } F(q) < 0, \text{ elastic deformation for all such q. Plastic otherwise.} \tag{1}$$

Naturally, the manifold given by $F(q) = 0$ represent the point of transition from elastic to plastic deformation. The manifold is known as the yield surface.

The specific $F(q)$ will depend on the model.

The complete characterisation of a plastic model requires the definition of the evolution laws for the internal variables.

$$\dot{q} = \dot{\lambda} Q(\sigma, q) \tag{2}$$

Where $\dot{\lambda}$ is a proportionality parameter called the plastic multiplier.

0.2 FreeFEM Implementation

Restarting this section entirely. 1st attempt did not work.

0.2.1 Physical Model

We concern ourselves with the model of a 50um by 75um block of copper impacting a rigid wall at 500m/s.

We proceed with the Johnson-Cook model of plasticity. [2]

The yield function of the Johnson-Cook plasticity model is given by the difference of the von Mises stress and the Johnson-Cook yield stress.

$$F(q) = \sigma_{vm} - \sigma_Y \tag{3}$$

$$\sigma_{vm} = \sqrt{\frac{3}{2} s : s} \ ; \ s = \sigma - \frac{1}{3} \operatorname{Tr}(\sigma) I \tag{4}$$

$$\sigma_Y = (A + B\bar{\epsilon}_p^{n_{JC}})(1 + C \ln(\dot{\bar{\epsilon}}_p / \dot{\epsilon}_0))(1 - \theta^m) \ ; \ \theta = \frac{T - T_0}{T_m - T_0} \tag{5}$$

Where σ_{vm} is the von Mises stress.

For the accumulation of plastic deformation here, it will be given by the flow rule and Drucker postulate of associative flow.

$$\dot{\epsilon}_p = \dot{\lambda} \frac{\partial F(q)}{\partial \sigma} = \dot{\lambda} \frac{3s}{2\sigma_{vm}} \tag{6}$$

$\dot{\lambda}$ is a proportionality parameter that needs to be determined. Fortunately for us, with von Mises' J_2 based Johnson-Cook associated flow plasticity, $\dot{\lambda}$ is equivalent to $\dot{\bar{\epsilon}}_p$.[1]

$$\dot{\bar{\epsilon}}_p = \sqrt{\frac{2}{3}\dot{\epsilon}_p : \dot{\epsilon}_p} = \sqrt{\dot{\lambda}^2 \frac{3s : s}{2\sigma_{vm}^2}} = \dot{\lambda} \tag{7}$$

As we will see later, we can enforce $\dot{\lambda} \geq 0$ to drop absolute value.

Johnson-Cook is also a thermomechanical model, so the power delivered via the plastic process is given by the following equation.

$$\dot{W}_p = \sigma : \dot{\epsilon}_p \tag{8}$$

We suppose only a fraction of the energy is turned into heat, so the temperature change is given by the following.

$$\dot{T} = \chi \frac{s : \dot{\epsilon}_p}{\rho C_p} = \chi \dot{\bar{\epsilon}}_p \frac{\sigma_{vm}}{\rho C_p} = T_k \dot{\bar{\epsilon}}_p \sigma_{vm} \tag{9}$$

Where we define $T_k = \frac{\chi}{\rho C_p}$ and call it the temp fraction.

We lay out all the constants used and move towards the numerical implementation.

```
// Copper particle
real rho         = 8960.;        // Density
real E           = 117e9;        // Young's modulus
real nu          = 0.34;         // Poisson modulus
real A           = 90e6;         // 1st Johnson-Cook
    hardening parameter
real B           = 292e6;        // 2nd Johnson-Cook
    hardening parameter
real nJC         = 0.31;         // Johnson-Cook hardening
    strain exponent
real C           = 0.025;        // Johnson-Cook rate
    parameter
real m           = 1.09;         // Johnson-Cook thermal
    exponent
real dEps0       = 1.0;          // Reference strain rate

real Tm          = 1356.;        // Melting temp
real Cp          = 385.;         // Specific heat
real chi         = 0.9;          // Heat fraction
real Tk          = chi/(rho*Cp); // Temp fraction
real T0     = 300.;              // Starting temp

real vimp   = -500.;     // Impact velocity
```

0.2.2 Elastic Portion only

We set up the model as a linear elastic problem, integrated over time to 50ns using Newark beta.

To avoid the cost of a contact algorithm, we idealize the circumstances and initiate the model right when the block hits the wall, and enforce a Dirichlet condition such that it doesn't penetrate it.

0.2.3 Plasticity with Linear Elasticity

We write everything out into incremental form.

We again use Newark beta to integrate, though this time the formula will be slightly different. The elastic internal stress term needs to be compensated with the existence of the plastic strain.

$$\sigma_{ij} = C_{ijkl}\epsilon_{ij} \Longrightarrow \sigma_{ij} = C_{ijkl}(\epsilon_{ij} - \epsilon_{ij}^{p}) \tag{10}$$

Giving us the differential equation in Newark's form, with M and K as defined in the contact pdf.

$$M\ddot{x} + Kx + C_{ijkl}\epsilon_{ij}^{p}(x)\epsilon_{kl}(v) = 0 \tag{11}$$

We now need to determine the equivalent plastic strain increment from equation 6. The determination of $\dot{\lambda}$ can be done numerically, and it is likely going to be some form of nonlinear optimization problem. We can apply Karush-Kuhn-Tucker (KKT) if we enforce a few criterions that are also sensible physics. KKT does the following with some restrictions.[5]

$$\text{minimize } f(x) \tag{12}$$

$$\text{subject to } g_i(x) \leq 0 \ ; \ h_j(x) = 0 \tag{13}$$

We can force $F(q) \leq 0$ and still maintain the flow rule and obtain the necessary restrictions for KKT to apply. Recall the assumptions in the yield criterion 1. If the process is elastic $-$ $F(q) < 0$, the plastic strain increment is 0, so in those cases $\dot{\lambda} = 0$. If the process is elastoplastic $-$ $F(q) = 0$ under the new restriction, the plastic strain increment is > 0, so $\dot{\lambda} > 0$. By enforcing a new condition $F(q) \leq 0$, we can obtain the needed extra h_j condition which we may notice is $F(q)\dot{\lambda} = 0$.

Giving us the following conditions for applying KKT.

$$\dot{\lambda} \geq 0 \tag{14}$$

$$F(q) \leq 0 \tag{15}$$

$$F(q)\dot{\lambda} = 0 \tag{16}$$

However, we will not be using KKT here, but the formulation above is general and can be applied to other plasticity models.

Recall in the model section that for von Mises' J_2 based associative plastic flow, $\dot{\lambda} = \dot{\bar{\epsilon}}_p$. With the condition that $F(q) = 0$ for plastic processes, it is much easier for us to simply Newton-Raphson solve for the root of $F(q)$ by updating $\dot{\bar{\epsilon}}_p$ in the Johnson-Cook yield stress. All that we have to be careful of is to maintain the restrictions at all times.

At the moment, I am feeling lazy, so we will currently adopt the much easier to implement secant method.

$$\dot{\lambda}_{n+1} = \dot{\lambda}_n - \frac{F(\dot{\lambda}_n)}{D_{\dot{\lambda}} F(\dot{\lambda}_n)} \tag{17}$$

$$D_{\dot{\lambda}} F(\dot{\lambda}_n) \approx \frac{F(\dot{\lambda}_n) - F(\dot{\lambda}_{n-1})}{\dot{\lambda}_n - \dot{\lambda}_{n-1}} \tag{18}$$

0.2.4 Results

We first present the elastic results. At the end of those 50ns, we already see the block begin to rebound off the wall. We are definitely accumulating errors due to not using a contact algorithm.

References

[1] Dassault Systèmes, Providence, RI. *Johnson-Cook Plasticity*, 2024. Abaqus 2024 Documentation.

[2] Gordon R. Johnson and William H. Cook. Fracture characteristics of three metals subjected to various strains, strain rates, temperatures and pressures. *Engineering Fracture Mechanics*, 21(1):31–48, 1985.

[3] Akhtar S. Khan and Sujian Huang. *Continuum Theory of Plasticity*. John Wiley & Sons, New York, 1995.

[4] Jacob Lubliner. *Plasticity Theory*. Macmillan, New York, 1990.

[5] Michaela Nagler, Astrid Pechstein, and Alexander Humer. A mixed finite element formulation for elastoplasticity. *International Journal for Numerical Methods in Engineering*, 123(21):5346–5368, 2022.